

NPI Doctor Notification System

Developer Technical Brief

Alert Physicians When Their NPI Is Used by a Vendor

ProMed Medical Supplies | Confidential | June 2026

CORE MISSION When a DME supplier, Home Health Agency, Hospice, or any other vendor submits a Medicare claim using a physician's NPI as the ordering/referring provider — the physician receives an immediate email or push notification showing exactly what was billed in their name. The doctor can then CONFIRM, DISPUTE, or REPORT the usage in one click. If rejected, the billing provider is notified and has 15 days to respond to Medicare directly or work with the physician to change their response.

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1. Why This App Is Needed

The problem physicians face today

The Problem

Every Medicare claim for DME, Home Health, or Hospice services requires an ordering or referring physician's NPI.

Vendors — including DME suppliers, home health agencies, and hospices — attach a physician's NPI to claims without the physician ever seeing it. This creates two major problems:

Problem	Who It Hurts	Current Gap
Fraudulent vendors forge or steal physician NPIs to bill Medicare for services never ordered	Physicians — their NPI tied to fraud they didn't commit	Physician has NO real-time visibility into who is using their NPI
Legitimate vendors bill under a physician's NPI without confirming the physician actually ordered the service	Medicare — paying for potentially unnecessary services	No automated notification system exists at the NPI level
Physicians only discover NPI misuse during an audit — often years later	Physicians — license risk, civil/criminal liability exposure	By the time fraud is found, millions may already be paid out
Ordering NPI is attached to claims for patients the physician never saw	Both physician and patient	No confirmation loop between vendor and ordering doctor

The Solution This App Provides

REAL-TIME NPI USAGE NOTIFICATION Every time a vendor (DME / Home Health / Hospice) submits or processes a Medicare claim using a physician's NPI as the ordering/referring provider, the physician immediately receives: **EMAIL:** Full claim details — what was billed, to which patient, by which vendor **PUSH NOTIFICATION:** Mobile alert with one-tap Confirm / Dispute action **PORTAL ENTRY:** Logged in physician's dashboard for full audit trail The physician responds. If they dispute, the flag is escalated to a compliance queue. If confirmed, the record is logged. If reported as fraud, it triggers a CMS/OIG referral.

2. How NPI Usage Works in Medicare Claims

Where physician NPIs appear and how to intercept them

2.1 Where a Physician's NPI Appears in a Claim

A physician's NPI can appear in multiple roles on a single Medicare claim. Each role is a trigger point for a notification.

NPI Role	Claim Field	Vendor Type	What It Means
Ordering Physician	Box 17 (CMS-1500) / NM1*DK (835)	DME, Lab, Imaging	Physician ordered the equipment or test
Referring Physician	Box 17 (CMS-1500) / NM1*DN (835)	Home Health, Hospice, Specialist	Physician referred the patient to the agency
Certifying Physician	Plan of Care / OASIS / NOE	Home Health, Hospice	Physician certified the patient meets eligibility criteria
Attending Physician	Hospice claim / NOE	Hospice	Primary physician overseeing hospice plan of care
Rendering Provider	Box 24J (CMS-1500)	Any provider type	Physician who actually performed the service

2.2 Vendor Types That Use Physician NPIs

Vendor Type	NPI Role Used	Medicare Claim Type	Frequency of Use
DME Supplier (e.g. ProMed)	Ordering Physician NPI	CMS-1500 / 837P	Every claim — required for all DME
Home Health Agency	Referring / Certifying Physician NPI	UB-04 / 837I	Every 30-day episode — OASIS + Plan of Care
Hospice	Attending / Certifying Physician NPI	UB-04 / 837I (NOE)	Admission + every 90-day recertification
Lab / Diagnostic	Ordering Physician NPI	CMS-1500 / 837P	Every test ordered
Skilled Nursing Facility	Attending Physician NPI	UB-04 / 837I	Per stay
Ambulance	Ordering Physician NPI	CMS-1500 / 837P	Per transport if physician-ordered

2.3 The 835 ERA — Where the App Intercepts the NPI

The 835 Electronic Remittance Advice (ERA) is the electronic file Medicare sends after processing a claim. It contains the ordering/referring NPI in structured EDI segments. This is the primary interception point for the notification system.

KEY 835 EDI SEGMENTS CONTAINING PHYSICIAN NPI: NM1*82*1*SMITH*JOHN****XX*1234567890 (Rendering Provider — NPI in position 9) NM1*DN*1*JONES*MARY****XX*9876543210 (Referring Provider NPI) NM1*DK*1*BROWN*ROBERT**XX*1122334455 (Ordering Provider NPI) NM1*77*2*ABC HOME HEALTH****XX*5544332211 (Service Facility NPI) The app parses every inbound 835 file and extracts all NPI values. Each extracted NPI is cross-referenced against the registered physician database. If a match is found → notification is triggered.

3. System Overview & User Flow

End-to-end data flow from claim submission to physician response

3.1 High-Level Data Flow

STEP 1 — CLAIM SUBMITTED Vendor (DME/HH/Hospice) submits claim to Medicare with physician NPI attached Source: 837 claim file via clearinghouse, or 835 ERA received from Medicare MAC STEP 2 — APP INGESTS CLAIM DATA App receives 835 ERA file (via webhook, SFTP, or clearinghouse API) Parser extracts: Ordering NPI, Referring NPI, Certifying NPI, Patient MBI, Vendor NPI, HCPCS codes, service dates, billed/paid amounts STEP 3 — NPI LOOKUP App queries physician registry: Is this NPI registered with our notification system? If YES → proceed to notification If NO → log the claim, flag for outreach to onboard physician STEP 4 — NOTIFICATION TRIGGERED Email sent to physician's registered address Push notification sent to physician's mobile app Entry created in physician's dashboard portal STEP 5 — PHYSICIAN RESPONDS CONFIRM: I ordered this. Record logged, no action. DISPUTE: I did not order this. Billing provider notified immediately. REPORT FRAUD: I never saw this patient. OIG referral triggered. STEP 5.5 — BILLING PROVIDER NOTIFIED ON REJECTION If physician disputes, the billing provider (vendor) is notified immediately Provider has 15 days to: (a) respond to Medicare directly, or (b) contact the physician and have them change their response to CONFIRM STEP 6 — RESOLUTION TRACKING All responses logged with timestamp Disputed/fraud cases escalated to compliance team if unresolved after 15 days Unresponded physician notifications auto-escalate after 72 hours

3.2 User Roles in the System

User Role	Who They Are	Primary Actions
Physician	MD/DO/NP/PA whose NPI is on the claim	Receive notifications, confirm/dispute/report, view history
Physician Office Admin	Staff authorized to act on behalf of physician	Same as physician — with physician consent during registration
Vendor (DME/HH/Hospice)	The company that submitted the claim	View confirmation status, receive dispute alerts, upload documentation
Compliance Officer	Internal or third-party compliance reviewer	Review disputed cases, escalate to OIG, generate reports
System Admin	App administrator	Manage physician accounts, monitor notification delivery, system health

4. Physician Registration & Profile

How doctors sign up and manage their notification preferences

4.1 Registration Flow

PHYSICIAN ONBOARDING STEPS: 1. Physician visits app portal (web or mobile) 2. Enters NPI number 3. System auto-populates name, specialty, address from NPPES registry 4. Physician verifies identity (options below) 5. Physician enters contact preferences (email, mobile, both) 6. Physician sets notification rules (see Section 5) 7. Account activated — monitoring begins immediately

4.2 Identity Verification Options

Method	How It Works	Security Level
DEA/License + NPI Match	Physician enters their state license number; system cross-checks NPI-to-license in NPPES	Medium
Email to NPI-Registered Address	System sends verification code to email address on file with NPPES	Medium
ID.me / Login.gov	Physician uses federal identity verification service (same as CMS provider portals)	High — Recommended
PECOS Credential Match	Physician enters PECOS enrollment credentials to verify Medicare provider status	High
Manual Verification	Physician uploads copy of DEA certificate + NPI confirmation letter; admin reviews	High — fallback method

4.3 Physician Profile Data Model

Field	Source	Required?	Notes
npi	Physician input	Yes	10-digit NPI — primary key
first_name / last_name	NPPES auto-populated	Yes	Editable by physician
specialty / taxonomy	NPPES auto-populated	Yes	Used to validate claim specialty match
practice_name	NPPES	Yes	
practice_address	NPPES	Yes	Used for geographic anomaly checks
email_primary	Physician input	Yes	All notifications sent here
email_secondary	Physician input	No	CC on all notifications
mobile_phone	Physician input	No	Required for push/SMS alerts
notification_prefs	Physician input	Yes	Email / Push / Both / Digest
alert_threshold	Physician input	Yes	All claims / Over \$X / New vendor only
office_admin_email	Physician input	No	Admin receives copy if set
verified	System	Yes	Boolean — must be true before notifications activate
verification_method	System	Yes	Tracks how identity was confirmed
registered_date	System	Yes	
last_login	System	Yes	

5. Trigger Engine — When to Send a Notification

Business rules that determine if and when a notification fires

5.1 Primary Trigger: NPI Match on Incoming Claim

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TRIGGER LOGIC (pseudocode): FUNCTION processIncomingClaim(claim): npis_to_check = [ claim.npi_ordering, claim.npi_referring, claim.npi_certifying, claim.npi_attending ] FOR each npi IN npis_to_check: IF npi IS NOT NULL: physician = lookupPhysician(npi) // query physician registry IF physician EXISTS AND physician.verified == TRUE: notification = buildNotification(claim, physician, npi_role) sendNotification(physician, notification) logNotificationRecord(notification)
```

5.2 Notification Rules — Physician Configurable

Physicians can customize when they receive notifications to avoid alert fatigue. The following rules are configurable per physician during onboarding.

Rule Setting	Options	Default	Description
Notify on every claim	Yes / No	Yes	Send notification for every single claim using their NPI
Notify for new vendors only	Yes / No	No	Only alert when a vendor they haven't seen before uses their NPI
Minimum claim amount	\$0 / \$100 / \$500 / \$1,000+	\$0 (all)	Only notify if claim amount exceeds threshold
Digest mode	Real-time / Daily / Weekly	Real-time	Bundle notifications into digest instead of instant alerts
Notify for specific vendor types	DME / HH / Hospice / All	All	Filter by vendor category
Notify admin copy	Yes / No	No	CC office administrator on every notification
Auto-confirm known vendors	Yes / No	No	Skip notification for pre-approved vendors physician has whitelisted
Escalate if no response in	24h / 48h / 72h / Never	72h	Auto-escalate to compliance if physician doesn't respond

5.3 Automatic Fraud Signal Escalation

Regardless of physician notification preferences, the following conditions automatically escalate to the compliance queue without waiting for physician response:

Condition	Auto-Action	Risk Level
Physician's NPI used by 3+ different vendors in the same week	Compliance alert + physician notification marked URGENT	HIGH
Claim amount > \$10,000 on first use by a new vendor	Compliance review queue + immediate physician notification	HIGH
NPI used for a patient in a different state than physician's practice	Geographic anomaly flag + physician notification	MEDIUM
Physician previously marked a vendor as disputed — same vendor bills again	Immediate compliance escalation + physician URGENT alert	CRITICAL

Vendor NPI is on OIG exclusion list	Block notification to physician — immediate compliance/OIG escalation	CRITICAL
Same patient billed for hospice + home health simultaneously	Dual-vendor conflict flag + both ordering physicians notified	HIGH

Exact content and format of physician notifications

FROM: alerts@npiwatch.com (or white-labeled domain) TO: dr.smith@medicalgroup.com SUBJECT: [NPI Alert] Your NPI was used on a Medicare claim — Action Required

[REDACTED] Your NPI (1234567890) was used on a Medicare claim [REDACTED] Dr. John Smith, A Medicare claim has been submitted using your NPI as the ordering/referring physician. Please review and confirm whether you authorized this.

CLAIM DETAILS

[REDACTED] Vendor: ABC Home Health Agency Vendor Type: Home Health Agency Vendor NPI: 9876543210 Vendor Address: 1234 Main St, Los Angeles, CA 90001 Patient: J*** S*** (last 4 of MBI: MK73) Date of Service: 06/01/2026 – 06/30/2026 Service Description: Home Health Services (Skilled Nursing, Physical Therapy) Amount Billed: \$3,850.00 Medicare Paid: \$3,250.00 Your Role on This Claim: REFERRING PHYSICIAN Claim Number: 987654321

[REDACTED] [CONFIRM — I authorized this] [DISPUTE — I did not order/refer this] [REPORT FRAUD — I never saw this patient] If you do not respond within 72 hours, this claim will be flagged for compliance review. Questions? Contact us at support@npiwatch.com or call 1-800-NPI-WATCH To manage your notification preferences, visit your portal.

Notification Field	Content
Title	NPI Alert — Action Required
Body (short)	ABC Home Health used your NPI for a \$3,250 claim. Tap to review.
Deep Link	Opens directly to claim detail screen in physician mobile app
Badge Count	Increments for each unreviewed notification
Actions (iOS/Android)	CONFIRM DISPUTE (available without opening app)
Sound	Default alert tone — urgent tone for fraud-flagged claims
Priority	High priority push (bypasses Do Not Disturb for CRITICAL flags)

SMS TEMPLATE (160 characters max): NPI ALERT: Your NPI was used by ABC Home Health for a \$3,250 Medicare claim. Reply CONFIRM, DISPUTE, or call 1-800-NPI-WATCH. Ref: CLM-987654321

7. Physician Response Workflow

What happens after the physician takes action

7.1 Response Options & Outcomes

Response	What Physician Is Saying	System Action	Vendor Notified?	CMS/OIG Action
CONFIRM	Yes, I ordered/referred this patient to this vendor	Claim marked CONFIRMED. Logged with timestamp and physician ID. No further action.	Yes — vendor sees green confirmation status	None
DISPUTE	I did not order or refer this. There may be an error.	Claim marked DISPUTED/REJECTED. Billing provider (vendor) immediately notified. Provider has 15 days to either respond to Medicare directly or contact the physician to have the response changed.	Yes — billing provider receives formal rejection notice with 15-day response window	Flagged for review if unresolved after 15 days
REPORT FRAUD	I never saw this patient. This is fraud.	Claim marked FRAUD REPORTED. Immediate compliance escalation. Case documented with all claim details. Attorney/OIG referral package auto-generated.	Yes — vendor receives fraud allegation notice (legal hold)	Auto-generated referral to OIG Hotline and CMS Medicare Fraud Strike Force
NO RESPONSE (72h)	Physician did not act	Auto-escalated to compliance queue. Second notification sent. Marked PENDING REVIEW.	No	Flagged if pattern of no-response on high-value claims

7.2 Dispute / Rejection Resolution Process

DISPUTE / REJECTION WORKFLOW TIMELINE: Day 0: Physician clicks DISPUTE (rejects the claim) → Billing provider (vendor) receives rejection notification IMMEDIATELY → Compliance case opened with case number → Physician receives case confirmation email → 15-day response clock starts for the billing provider Days 1-15: Billing Provider Response Window (15 Days) The billing provider has two options during this window: OPTION A — Respond to Medicare directly: → Provider submits documentation/appeal directly through Medicare → Provider uploads proof of response into the app (claim number, date submitted to Medicare, supporting documentation) OPTION B — Contact the physician to change their response: → Provider reaches out to physician's office to clarify/correct the order → If physician agrees, physician can CHANGE their response in the portal from DISPUTE back to CONFIRM (with a logged reason/note) Day 15: If billing provider responded to Medicare: → Case marked RESPONDED_TO_MEDICARE, routed to compliance for outcome tracking If physician changed their response to CONFIRM: → Case marked RESOLVED_BY_PHYSICIAN, claim notification updated to CONFIRMED If NEITHER occurs within 15 days: → Case auto-escalates to CMS/compliance for payment review and potential recoupment action — billing provider marked NON-RESPONSIVE Day 30: Unresolved disputes (no Medicare response, no physician status change) flagged for CMS RAC audit referral

7.3 Billing Provider Notification on Rejection

When a physician disputes/rejects a claim, the billing provider (the vendor who submitted the claim — DME supplier, Home Health Agency, Hospice, etc.) must be notified immediately, not just logged internally. This mirrors the physician notification system in reverse.

BILLING PROVIDER REJECTION NOTICE — EMAIL TEMPLATE SUBJECT: [URGENT] Physician Rejected Claim #987654321 — 15-Day Response Required ABC Home Health Agency, Dr. John Smith (NPI 1234567890) has **REJECTED** the following claim submitted using their NPI as the referring physician: Claim Number: 987654321 Patient: J*** S*** (MBI ending MK73) Date of Service: 06/01/2026 – 06/30/2026 Amount Billed: \$3,850.00 Reason Given: [physician's optional note, if provided] **YOU HAVE 15 DAYS FROM TODAY (by 07/01/2026) TO:** 1. Respond to Medicare directly with supporting documentation, **OR** 2. Contact Dr. Smith's office to resolve the discrepancy — if resolved, the physician must update their response in the portal to **CONFIRM** If neither action is taken within 15 days, this claim will be escalated to CMS compliance for payment review and possible recoupment. [Upload Medicare Response Documentation] [View Case Details]

Field	Description
Trigger	Physician selects DISPUTE on a notification
Recipient	Billing provider's registered compliance/billing contact email
Response Window	15 calendar days from rejection timestamp
Option A	Provider responds to Medicare directly — must log proof of submission in-app
Option B	Provider contacts physician — physician may change response from DISPUTE to CONFIRM
Reminder Cadence	Day 7 reminder, Day 13 final reminder (2 days before deadline)
Day 15 Outcome (resolved)	Case closed as RESPONDED_TO_MEDICARE or RESOLVED_BY_PHYSICIAN
Day 15 Outcome (unresolved)	Auto-escalated to CMS/compliance — provider marked NON-RESPONSIVE

7.4 Physician Response Change Flow

If the billing provider successfully clarifies the order with the physician's office during the 15-day window, the physician must be able to formally change their response. This is logged as a distinct event — not a silent edit.

PSEUDOCODE — changeResponse(): FUNCTION changeResponse(notification_id, new_response, physician_npi, reason): IF notification.status != 'DISPUTED': REJECT // can only change a response that is currently disputed IF daysSinceDispute(notification_id) > 15: REJECT // window has closed — must go through compliance instead UPDATE notification SET status = new_response, // typically 'CONFIRMED' response_changed = TRUE, response_change_reason = reason, response_changed_at = NOW() NOTIFY billing_provider: 'Dr. Smith has updated their response to CONFIRMED' CLOSE associated DISPUTE_CASE with resolution_notes = reason

8. Vendor-Side Visibility

What DME suppliers, home health agencies, and hospices see

8.1 Vendor Dashboard

Vendors who register with the platform can see real-time confirmation status for their claims. This incentivizes them to use the platform proactively to prove legitimacy.

Dashboard Element	What Vendor Sees
Claim Status Feed	All recently submitted claims with physician NPI confirmation status: CONFIRMED / PENDING / DISPUTED / FRAUD REPORTED
Physician Response Rate	Percentage of their claims that have been confirmed — a trust score visible to compliance reviewers
Dispute Center	All active disputes with deadline counter, document upload, and response tools
Pre-Authorization Tool	Before submitting a claim, vendor can send a pre-authorization request to the physician for sign-off
Physician Contact Request	If physician is not registered, vendor can request system to send onboarding invite to physician
Audit Package Generator	For any confirmed claim, generate a compliance package with physician confirmation, claim details, and documentation

8.2 Pre-Authorization Feature (Optional Add-On)

This is a proactive feature: instead of waiting for a claim to be filed, vendors can request pre-authorization from the physician BEFORE billing Medicare.

PRE-AUTHORIZATION FLOW: 1. Vendor initiates pre-auth request in app: Patient: John Smith | Service: Power Wheelchair K0823 | Est. Cost: \$4,200 2. Physician receives notification: 'ABC DME is requesting authorization to bill Medicare for a power wheelchair for your patient John Smith. Please approve or deny.' 3. Physician responds: APPROVE or DENY 4. If APPROVED: pre-auth token generated — attached to claim when filed Pre-authorized claims automatically CONFIRMED when ERA is received 5. If DENIED: vendor cannot proceed with that patient/service combination Denial logged — if vendor bills anyway, immediate FRAUD flag triggered

9. Database Schema

Core tables for the notification system

Table: PHYSICIAN

npi VARCHAR(10) PRIMARY KEY first_name VARCHAR(100) NOT NULL last_name VARCHAR(100) NOT NULL specialty VARCHAR(200) taxonomy_code VARCHAR(10) practice_name VARCHAR(300) practice_address JSONB email_primary VARCHAR(200) NOT NULL email_secondary VARCHAR(200) mobile_phone VARCHAR(20) notification_mode VARCHAR(20) -- REALTIME, DAILY, WEEKLY alert_threshold DECIMAL(10,2) DEFAULT 0 vendor_type_filter JSONB -- ['DME','HOME_HEALTH','HOSPICE'] whitelisted_vendors JSONB -- Array of pre-approved vendor NPIs auto_escalate_hours INT DEFAULT 72 verified BOOLEAN DEFAULT FALSE verification_method VARCHAR(50) registered_date TIMESTAMP DEFAULT NOW() last_login TIMESTAMP

Table: CLAIM_NOTIFICATION

notification_id SERIAL PRIMARY KEY claim_number VARCHAR(30) NOT NULL physician_npi VARCHAR(10) REFERENCES PHYSICIAN(npi) physician_npi_role VARCHAR(30) -- ORDERING, REFERRING, CERTIFYING, ATTENDING vendor_npi VARCHAR(10) NOT NULL vendor_name VARCHAR(300) vendor_type VARCHAR(50) -- DME, HOME_HEALTH, HOSPICE, etc. patient_mbi VARCHAR(11) patient_name_partial VARCHAR(50) -- Masked: J*** S*** dos_from DATE dos_to DATE service_description TEXT hcpcs_codes JSONB amount_billed DECIMAL(12,2) amount_paid DECIMAL(12,2) era_file_id VARCHAR(50) notification_sent_at TIMESTAMP email_sent BOOLEAN DEFAULT FALSE push_sent BOOLEAN DEFAULT FALSE sms_sent BOOLEAN DEFAULT FALSE status VARCHAR(20) -- PENDING, CONFIRMED, DISPUTED, FRAUD_REPORTED, ESCALATED physician_response VARCHAR(20) response_at TIMESTAMP response_changed BOOLEAN DEFAULT FALSE -- physician changed DISPUTE to CONFIRM response_change_reason TEXT response_changed_at TIMESTAMP escalated_at TIMESTAMP created_at TIMESTAMP DEFAULT NOW()

Table: DISPUTE_CASE

case_id SERIAL PRIMARY KEY notification_id INT REFERENCES CLAIM_NOTIFICATION(notification_id) physician_npi VARCHAR(10) REFERENCES PHYSICIAN(npi) vendor_npi VARCHAR(10) dispute_type VARCHAR(20) -- DISPUTE, FRAUD_REPORT physician_notes TEXT billing_provider_notified_at TIMESTAMP -- rejection notice sent to vendor response_due_date DATE -- rejection_date + 15 days provider_response_type VARCHAR(30) -- RESPONDED_TO_MEDICARE, PHYSICIAN_CHANGED_RESPONSE, NONE medicare_response_proof JSONB -- claim #, date submitted, docs uploaded by provider vendor_response TEXT vendor_docs JSONB -- Array of uploaded document URLs vendor_responded_at TIMESTAMP reminder_sent_day7 BOOLEAN DEFAULT FALSE reminder_sent_day13 BOOLEAN DEFAULT FALSE status VARCHAR(20) -- OPEN, RESPONDED_TO_MEDICARE, RESOLVED_BY_PHYSICIAN, NON_RESPONSIVE, CLOSED, REFERRED_OIG compliance_reviewer VARCHAR(100) oig_referral_date DATE oig_case_number VARCHAR(50) opened_at TIMESTAMP DEFAULT NOW() closed_at TIMESTAMP resolution_notes TEXT

Table: PREAUTH_REQUEST

preauth_id SERIAL PRIMARY KEY physician_npi VARCHAR(10) REFERENCES PHYSICIAN(npi) vendor_npi VARCHAR(10) patient_mbi VARCHAR(11) patient_name VARCHAR(200) service_description TEXT hcpcs_code VARCHAR(10) estimated_amount DECIMAL(12,2) status VARCHAR(20) -- PENDING, APPROVED, DENIED physician_response VARCHAR(10) response_at TIMESTAMP preauth_token VARCHAR(50) UNIQUE -- Used on actual claim expires_at TIMESTAMP created_at TIMESTAMP DEFAULT NOW()

10. API Endpoints

Full REST API spec for app development

10.1 Physician Endpoints

Method	Endpoint	Description
POST	/api/v1/physicians/register	Register new physician with NPI
POST	/api/v1/physicians/verify	Submit verification credentials
GET	/api/v1/physicians/{npi}/profile	Get physician profile
PUT	/api/v1/physicians/{npi}/profile	Update contact info and preferences
PUT	/api/v1/physicians/{npi}/whitelist	Add/remove pre-approved vendors
GET	/api/v1/physicians/{npi}/notifications	List all notifications (paginated)
GET	/api/v1/physicians/{npi}/notifications/stats	Summary counts by status

10.2 Notification Endpoints

Method	Endpoint	Description
GET	/api/v1/notifications/{id}	Get notification detail
POST	/api/v1/notifications/{id}/respond	Physician response: CONFIRM/DISPUTE/FRAUD
POST	/api/v1/notifications/{id}/resend	Resend notification to physician
GET	/api/v1/notifications/pending	List all pending (unresponded) notifications
POST	/api/v1/claims/ingest	Ingest 835 ERA file — triggers notifications
POST	/api/v1/claims/ingest/single	Ingest single claim JSON — triggers notification
GET	/api/v1/claims/{claim_number}	Get claim with notification status

10.3 Dispute & Case Endpoints

Method	Endpoint	Description
GET	/api/v1/disputes	List all dispute cases (filterable)
GET	/api/v1/disputes/{case_id}	Get dispute case detail
POST	/api/v1/disputes/{case_id}/vendor-response	Vendor submits documentation
POST	/api/v1/disputes/{case_id}/medicare-response	Provider logs proof they responded to Medicare directly (claim #, date, docs)
POST	/api/v1/notifications/{id}/change-response	Physician changes DISPUTE to CONFIRM within the 15-day window
GET	/api/v1/disputes/{case_id}/deadline	Get days remaining on the 15-day provider response window
POST	/api/v1/disputes/{case_id}/close	Compliance closes a case
POST	/api/v1/disputes/{case_id}/refer-oig	Generate and send OIG referral
GET	/api/v1/disputes/{case_id}/oig-package	Download OIG referral package PDF

10.4 Pre-Authorization Endpoints

Method	Endpoint	Description
POST	/api/v1/preauth/request	Vendor creates pre-auth request
GET	/api/v1/preauth/{preauth_id}	Get pre-auth status
POST	/api/v1/preauth/{preauth_id}/respond	Physician approves or denies
GET	/api/v1/preauth/validate/{token}	Validate pre-auth token on claim submission

10.5 Webhook Endpoints (for ERA/Clearinghouse)

Method	Endpoint	Description
POST	/api/v1/webhooks/era	Receive 835 ERA file from clearinghouse
POST	/api/v1/webhooks/claim-status	Receive claim status updates (277)
GET	/api/v1/webhooks/health	Webhook health check endpoint

11. Integration Data Sources

External systems the app must connect to

Integration	URL / Source	Frequency	Purpose	Auth Required
NPPES NPI Registry	npiregistry.cms.hhs.gov/api	Real-time	Validate NPI, pull physician name/specialty/address for auto-populate	None — public API
OIG LEIE Exclusion List	oig.hhs.gov/exclusions	Monthly sync	Check if vendor NPI is excluded before sending physician notification	None — public download
CMS PECOS	pecos.cms.gov	Weekly	Verify physician has active Medicare enrollment	CMS credentials
Medicare 835 ERA	Via clearinghouse (Availity, Change Healthcare, Waystar)	Daily / real-time	Primary claim data source — triggers all notifications	Clearinghouse API key
Email Service	SendGrid / AWS SES	Real-time	Deliver email notifications to physicians	API key
Push Notifications	Firebase FCM / Apple APNs	Real-time	Deliver mobile push alerts	Firebase / Apple certs
SMS Gateway	Twilio / AWS SNS	Real-time	Deliver SMS fallback notifications	Twilio API key
Identity Verification	ID.me or Login.gov	On registration	Verify physician identity during onboarding	OAuth 2.0
OIG Fraud Hotline	oig.hhs.gov/fraud	On-demand	Submit fraud reports when physician clicks REPORT FRAUD	API or web form

12. UI/UX Screen Specifications

Required screens for web portal and mobile app

Screen 1: Physician Registration / Onboarding

Platform: Web portal

- Step 1 — Enter NPI: large input field, auto-lookup pulls name/specialty from NPPES
- Step 2 — Verify Identity: choice of ID.me, PECOS, or license number
- Step 3 — Contact Preferences: email(s), mobile, notification mode
- Step 4 — Notification Rules: vendor types, amount threshold, digest vs real-time
- Step 5 — Review & Activate: summary of settings before going live

Screen 2: Physician Dashboard (Home)

Platform: Web + Mobile

- Notification counter by status: Pending / Confirmed / Disputed / Fraud Reported
- Recent activity feed: last 10 notifications with vendor name, amount, status badge
- Quick stats: total claims using your NPI this month, total confirmed, total disputed
- Alert banner for any URGENT or CRITICAL flags requiring immediate attention
- Quick action button: Review Pending Notifications

Screen 3: Notification Detail — Claim Review

Platform: Web + Mobile

- Vendor info card: name, type, NPI, address, Medicare enrollment status
- Claim info card: patient (masked), dates of service, service description, HCPCS codes
- Amount block: billed / approved / paid — shown clearly
- Your role on this claim (Ordering / Referring / Certifying / Attending)
- Action buttons: CONFIRM (green) | DISPUTE (orange) | REPORT FRAUD (red)
- Note field: optional — physician can add notes before responding
- Timer: shows time remaining before auto-escalation

Screen 4: Dispute Case Management

Platform: Web portal

- Open cases list with days-remaining counter on vendor response deadline
- Case detail: full claim info + physician dispute notes + vendor uploaded documents
- Document viewer: inline PDF/image viewer for vendor-uploaded documentation
- Resolution actions: Accept Vendor Docs (close case) / Maintain Dispute / Escalate
- OIG Referral button: one-click generates referral package PDF

Screen 5: Notification History

Platform: Web + Mobile

- Full searchable list of all past notifications
- Filter by: vendor name, vendor type, date range, status, amount
- Export: CSV or PDF export of notification history for personal records
- Audit trail: full log of all actions taken on each notification

Screen 6: Vendor Pre-Auth Portal (Separate Vendor Login)

Platform: Web portal

- Claim status board: all active claims with physician confirmation status
- Pre-auth request form: patient, service, HCPCS, estimated amount
- Dispute response center: upload documentation for disputed claims
- Confirmation rate score: visible metric showing their legitimacy standing

13. Compliance, HIPAA & Legal Considerations

Critical legal framework your developer and legal team must address

13.1 HIPAA Compliance

- PHI displayed in notifications must be minimized — patient name should be masked (J*** S***) in email/push; full name only in secure portal after login
- All data encrypted at rest (AES-256) and in transit (TLS 1.3)
- Physician portal requires MFA — PHI is accessible only after authenticated session
- Audit logs of all PHI access must be maintained for 6 years
- BAA required with all vendors: SendGrid, Twilio, Firebase, cloud host, clearinghouse
- Patient MBI must never appear in email subject lines or push notification previews

13.2 IMPORTANT — Legal Framework Notes

CRITICAL LEGAL ITEMS — Review with Healthcare Attorney Before Launch: 1. This app does NOT have direct access to Medicare claims data from CMS. It processes 835 ERA files that PROVIDERS/VENDORS receive and share with the app. Physicians and vendors must authorize data sharing via signed consent. 2. A physician's response (CONFIRM / DISPUTE / REPORT FRAUD) is not a legal determination of fraud. It is an attestation of their knowledge at the time. All fraud reports must be clearly labeled as 'alleged' until investigated. 3. OIG referrals must follow 42 CFR Part 455 procedures. Auto-generated OIG referral packages must be reviewed by compliance staff before submission — do NOT auto-submit without human review. 4. Anti-Kickback Statute (AKS): Ensure the notification/confirmation system cannot be used to create improper financial relationships between physicians and vendors (e.g., vendors cannot pay physicians for confirmations). 5. False Claims Act exposure: Disputed claims that are paid must be reported to CMS under the 60-day repayment rule if fraud is confirmed.

13.3 Physician Privacy Protections

- Physicians' email addresses and mobile numbers are NEVER shared with vendors
- Vendors only see the confirmation status (Confirmed / Pending / Disputed) — not physician contact info
- Physicians can deactivate their account at any time — historical data retained per HIPAA
- Physician consent required before any data is used for analytics or reporting

14. Tech Stack & Development Milestones

Recommended technologies and build timeline

14.1 Recommended Tech Stack

Layer	Recommended Tool	Purpose
Backend API	Python / FastAPI or Node.js / Express	Core API, 835 parser, notification triggers
835 EDI Parser	pyx12 (Python) or custom Node parser	Parse incoming ERA files to structured data
Database	PostgreSQL 15+	Claims, notifications, disputes, physicians
Email Delivery	SendGrid or AWS SES	HIPAA-compliant email notifications
Push Notifications	Firebase FCM + Apple APNs	iOS and Android push alerts
SMS	Twilio	SMS fallback notifications
Job Queue	Redis + Celery or Bull (Node)	Async notification processing
Identity Verification	ID.me or Login.gov OAuth	Physician identity verification
Mobile App	React Native or Flutter	iOS + Android physician app
Web Portal	React + TypeScript	Physician and vendor web dashboards
Auth	Auth0 or AWS Cognito	MFA, JWT, role-based access
File Storage	AWS S3	835 files, vendor documents, HIPAA BAA
Hosting	AWS or Azure (HIPAA BAA available)	HIPAA-compliant cloud hosting
Monitoring	Datadog or AWS CloudWatch	Alert on notification failures

14.2 Development Milestones

Phase	Timeline	Deliverables
Phase 1: Foundation	Weeks 1–3	Database schema, NPPES NPI lookup API, physician registration + verification flow, basic auth
Phase 2: 835 Parser + Trigger Engine	Weeks 4–6	835 ERA file parser, NPI extraction, match-to-physician logic, notification queue
Phase 3: Email + Push Notifications	Weeks 7–8	SendGrid email templates, Firebase push setup, SMS via Twilio, notification delivery tracking
Phase 4: Physician Web Portal	Weeks 9–11	Dashboard, notification detail screen, confirm/dispute/fraud response flow, notification history
Phase 5: Mobile App	Weeks 12–14	React Native iOS + Android app, push notification integration, one-tap response actions
Phase 6: Dispute & Case Management	Weeks 15–16	Dispute workflow, vendor document upload, OIG referral package generator
Phase 7: Vendor Portal	Weeks 17–18	Vendor dashboard, claim confirmation status board, pre-auth request flow
Phase 8: Testing + HIPAA Audit	Weeks 19–20	Penetration testing, HIPAA security review, BAA execution with all vendors
Phase 9: Pilot Launch	Week 21–22	Soft launch with 50–100 test physicians, refine notification templates, fix issues

Phase 10: Full Launch	Week 23–24	Full production launch, physician onboarding campaign, monitoring active
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MVP IN 8 WEEKS (Phases 1-3): Physician registers with NPI App parses 835 ERA files Physician receives email when their NPI is used on a claim Physician can CONFIRM or DISPUTE via email link This is the core loop — everything else is an enhancement built on top of it.